

COMP47670 Assignment 1 - Autumn 2026/27

Deadline: Friday 30 October 2026

Overview

The objective of this assignment is to extract a dataset from a collection of web pages and use Python to prepare, analyse, and derive insights from the collected data.

You must complete the assignment as **three Jupyter notebooks**, with one notebook for each task described below. Script files should not be submitted. Each notebook must be clearly documented using code comments and Markdown cells. All discussion relating to a task must appear in Markdown cells in the corresponding notebook rather than in a separate document. Your discussion should explain and justify the decisions you made, with reference to the specifics of your own dataset, code, and results.

The four data sources for this assignment can be accessed at:

<http://mlg.ucd.ie/modules/python/assign1>

Task 1: Data Collection

- a) Choose **one** of the four provided data sources. Use Python web-scraping techniques to collect and parse all data from your chosen source. Save the resulting raw dataset in an appropriate format so that it can be loaded and processed in Task 2.
- b) In Markdown cells in this notebook, describe your approach to collecting and parsing the data, discuss any challenges you encountered, and explain the decisions you made during the data-collection process.

Task 2: Data Preprocessing

- a) Load the raw dataset saved in Task 1 into an appropriate data structure to build your Analytics Base Table (ABT). Apply the preprocessing steps needed to clean, filter, and transform it for analysis. Save the cleaned ABT in an appropriate format for use in Task 3.
- b) In Markdown cells in this notebook, describe and justify your preprocessing decisions, identify specific data-quality issues found in your dataset, explain how you addressed those issues, and explain why you selected your chosen methods over any alternatives you considered.

Task 3: Data Analysis

- a) Load the cleaned ABT created in Task 2. Analyse, characterise, and summarise the dataset, using appropriate statistics and visualisations to support your analysis.
- b) In Markdown cells in this notebook, interpret your results with reference to specific statistics, tables, or figures that you produced, summarise the main insights gained from your analysis, and suggest relevant ideas for further analysis or future work.

Use of Generative AI

- Generative AI tools, code assistants, and large language models (such as ChatGPT) **must not** be used for any part of this assignment, including in the discussion sections.

Guidelines

- The assignment should be completed **individually**. All submissions will be subject to plagiarism checking. Any evidence of plagiarism can result in a 0 grade.
- The grade awarded will be based on the complexity and appropriateness of the technical work and the quality and depth of the accompanying discussion.
- All discussion and interpretation must be grounded in your own results. Support observations with specific evidence from your analysis. Generic descriptions of methods are not sufficient.
- See the associated Grading Rubric document for a detailed breakdown of marks for each task.
- Submit your assignment via Brightspace as **a single ZIP file** containing:
 1. Your three Jupyter notebooks (IPYNB files, not HTML).
 2. The raw dataset saved in Task 1.
 3. The clean dataset saved in Task 2.
- The UCD *Additional Considerations for Assessment* policy applies for this module:
 - Students can request an assignment extension of up to 5 working days via InfoHub without academic approval.
 - With approval, this can be extended to an additional 5 days (i.e., 10 working days) with a one-grade-point deduction (e.g B+ to B).
 - Any further extension requires a certified application.